

PORTO SANTO AP, Portugal

WMO: 085240

Lat: 33.0714N

Lon: 16.3486W

Elev: 83

StdP: 100.33

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.9	12.8	5.1	5.5	14.6	6.0	5.9	14.9	12.9	15.5	11.7	15.4	3.3	340	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	26.1	21.4	25.2	20.8	24.9	20.6	22.6	24.6	21.9	24.2	21.4	23.7	4.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	16.8	23.7	21.1	15.9	23.2	20.5	15.3	22.9	66.6	24.4	64.3	24.4	62.4	23.6	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.4	8.5	DB	9.9	28.8	1.0	2.2	9.2	30.4	8.6	31.7	8.0	33.0	7.3	34.6	(4)
(5)			WB	7.8	23.2	0.8	1.0	7.2	23.9	6.7	24.5	6.3	25.0	5.7	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	16.1	15.8	16.3	16.9	18.3	20.3	21.8	22.9	22.6	21.2	18.9	17.3	(6)
(7)		DBStd	2.80	1.34	1.23	1.35	1.16	1.22	1.12	1.04	0.86	1.03	1.23	1.55	1.34	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	316	71	71	66	44	15	1	0	0	0	1	11	38	(9)
(10)		CDD10.0	3305	188	162	194	208	258	308	367	401	377	348	267	226	(10)
(11)		CDD18.3	579	1	1	2	2	14	58	109	142	127	90	28	5	(11)
(12)		CDH23.3	742	0	0	0	2	4	20	117	294	236	66	3	0	(12)
(13)		CDH26.7	21	0	0	0	0	0	1	6	7	7	1	0	0	(13)
(14)	Wind	WSAvg	4.7	4.6	4.8	5.0	4.9	4.9	4.8	4.9	4.6	4.1	4.3	4.7	4.7	(14)
(15)	Precipitation	PrecAvg	376	54	43	44	24	13	8	3	5	18	50	54	64	(15)
(16)		PrecMax	597	181	112	127	58	68	33	13	14	70	158	177	203	(16)
(17)		PrecMin	205	7	0	2	3	0	0	0	0	0	2	6	10	(17)
(18)		PrecStd	96	48	28	33	14	13	7	3	4	15	43	39	45	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	20.2	21.4	22.2	23.5	25.0	26.9	27.2	27.2	25.9	23.9	21.5	(19)
(20)			MCWB	17.3	16.6	16.3	16.9	18.6	20.5	20.6	21.7	21.5	21.1	20.1	18.1	(20)
(21)		2%	DB	19.3	19.2	20.1	20.8	22.1	23.9	25.2	26.1	26.0	24.8	22.9	20.8	(21)
(22)			MCWB	16.7	16.1	16.4	16.8	18.0	19.7	20.6	21.5	21.5	20.6	19.7	17.8	(22)
(23)		5%	DB	18.9	18.7	19.2	19.9	21.2	23.1	24.4	25.2	25.2	24.0	22.0	20.1	(23)
(24)			MCWB	16.3	15.7	15.9	16.2	17.2	19.0	20.0	20.8	20.8	20.0	18.8	17.2	(24)
(25)		10%	DB	18.2	18.0	18.6	19.1	20.8	22.3	23.9	24.9	24.8	23.2	21.2	19.3	(25)
(26)			MCWB	15.6	15.1	15.5	15.5	16.9	18.4	19.6	20.6	20.5	19.5	18.1	16.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	17.8	18.4	18.2	19.6	21.3	22.5	23.4	23.8	22.5	21.4	19.4	(27)
(28)			MCDWB	19.4	19.3	19.7	20.5	22.2	23.7	24.8	25.4	24.9	24.1	22.7	20.5	(28)
(29)		2%	WB	17.7	17.0	17.5	17.4	18.6	20.5	21.6	22.5	22.7	21.7	20.5	18.6	(29)
(30)			MCDWB	18.8	18.7	19.1	19.7	21.0	22.8	23.9	24.8	24.5	23.5	22.0	20.0	(30)
(31)		5%	WB	17.0	16.4	16.7	16.8	18.0	19.8	21.0	21.9	21.9	21.1	19.6	17.9	(31)
(32)			MCDWB	18.4	18.1	18.5	19.0	20.4	22.1	23.3	24.4	24.1	23.0	21.3	19.4	(32)
(33)		10%	WB	16.3	15.6	16.1	16.2	17.4	19.3	20.4	21.2	21.2	20.6	18.9	17.4	(33)
(34)			MCDWB	17.8	17.4	17.9	18.5	19.9	21.7	23.0	23.9	23.6	22.6	20.7	19.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.8	4.0	4.3	4.6	4.5	4.4	4.4	4.5	4.4	3.9	3.8	(35)	
(36)			MCDBR	4.3	4.8	4.9	5.3	5.0	4.8	4.9	4.8	5.0	4.9	4.4	4.2	(36)
(37)			MCWBR	3.3	3.5	3.3	3.3	3.0	2.8	2.8	2.7	2.9	3.0	3.1	3.1	(37)
(38)		5% WB	MCDBR	3.9	4.5	4.6	5.1	4.9	4.6	4.9	4.7	4.9	4.5	4.0	3.8	(38)
(39)			MCWBR	3.5	3.6	3.5	3.6	3.1	2.9	3.1	3.0	3.3	3.2	3.3	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.358	0.384	0.375	0.379	0.379	0.387	0.390	0.387	0.372	0.362	0.343	(40)	
(41)		taud	2.562	2.487	2.406	2.448	2.461	2.483	2.466	2.476	2.501	2.534	2.523	2.555	(41)	
(42)		Ebn at Noon	881	895	893	914	908	902	894	888	879	868	844	852	(42)	
(43)		Edh at Noon	82	97	113	113	112	110	111	108	102	91	85	79	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.75	3.59	4.78	5.95	6.44	6.85	6.76	6.39	5.44	4.17	2.94	2.49	(44)	
(45)		RadStd	0.13	0.27	0.26	0.23	0.47	0.40	0.38	0.32	0.19	0.17	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page